

Predicting Answer Correctness from Token-Level Top-10 Probability Curves

Özgür Aslan

Yildiz Technical University

Computational Semantics Final Project Report

Abstract—This report studies whether a language model’s token-level uncertainty can predict the correctness of its mathematical answers. For each Turkish GSM8K question, Qwen3-4B was sampled ten times and the sum of the probabilities of the ten most likely next tokens was recorded at every generated position. The resulting curve is close to one for most tokens, but small drops indicate that probability mass is spreading beyond the top candidates. Only mixed questions, where the same prompt produced both correct and incorrect answers, were kept so that the classifier must distinguish answer quality rather than question difficulty. The final dataset contains 1,320 answers from 132 mixed questions. Curve statistics, dip features, answer length, and within-question relative features were used with several tree-based classifiers. The best ROC-AUC was 0.6151 with Gradient Boosting, while Random Forest reached the best accuracy of 0.5781 against a 0.5250 majority baseline. The results show that the top-10 probability curve carries a real but weak correctness signal.

I. INTRODUCTION

Reasoning models often produce several plausible solution paths for the same mathematical problem. In a reinforcement-learning setting such as GRPO, it is useful to understand whether incorrect answers leave a detectable trace before the final answer is emitted. This project tests a simple hypothesis: when the model is about to choose a wrong reasoning branch, its next-token distribution becomes less concentrated, and this hesitation appears as a drop in the summed probability of the top-10 tokens.

The task is intentionally harder than ordinary answer classification. The input to the classifier is not the generated text, the question, or the final numeric answer. It only receives the graph-derived signal extracted during generation. Furthermore, the dataset keeps only questions for which the same model produced both correct and incorrect samples. This prevents a classifier from simply learning that some questions are easy while others are hard.

The project repository is available at <https://github.com/ozguraslank/ytu-nlp-project>.

II. DATA GENERATION

The source questions are Turkish grade-school math problems from local CSV samples derived from `ytu-ce-cosmos/gsm8k_tr`, a Turkish GSM8K dataset hosted on Hugging Face [2] and based on GSM8K [1]. The experiment did not run over the full Turkish GSM8K dataset. Instead, the generation script loads 250 training and 250 test questions, prompts `unsloth/Qwen3-4B` with a Turkish

TABLE I
FINAL DATASET AFTER FILTERING TO MIXED QUESTIONS.

Split	Questions	Answers	Correct	Incorrect
Train	68	680	376	304
Test	64	640	336	304
Total	132	1320	712	608

math assistant instruction, disables Qwen’s explicit thinking blocks, and samples ten answers per question using vLLM. Sampling used temperature 0.7, top-p 0.8, a maximum of 1024 generated tokens, and top-10 log-probability output for every generated token.

For token position t , vLLM returns log-probabilities for the highest scoring tokens. The signal used in this project is

$$s_t = \sum_{k=1}^{10} \exp(\ell_{t,k}), \quad (1)$$

where $\ell_{t,k}$ is the k th largest log-probability at position t . In a highly concentrated distribution, s_t is close to one. Lower values mean that more probability mass has moved outside the ten most likely tokens.

Each answer was labeled by parsing the numeric result after the required `####` marker, with a fallback to the last number in the answer text. Turkish decimal and thousands separators were normalized before comparison with the gold answer. A question was kept only if at least one of its ten samples was correct and at least one was incorrect.

Table I summarizes the final data. Among the 500 sampled prompts (250 train and 250 test), only 132 questions produced mixed answer sets containing both labels. The split is still by question, so sibling answers of the same question never appear in both training and test data.

III. TOP-10 PROBABILITY CURVES

Figure 1 shows one mixed question with five correct and five incorrect answers. The y-axis is deliberately zoomed because the raw top-10 probability mass is saturated near one. On a full 0–1 axis, almost all curves would appear flat. In this example, several incorrect answers show deeper and longer drops than most correct answers, especially in the middle and later parts of the generation.

A single question is only anecdotal, so the grid in Fig. 2 compares several mixed questions. The pattern is heteroge-

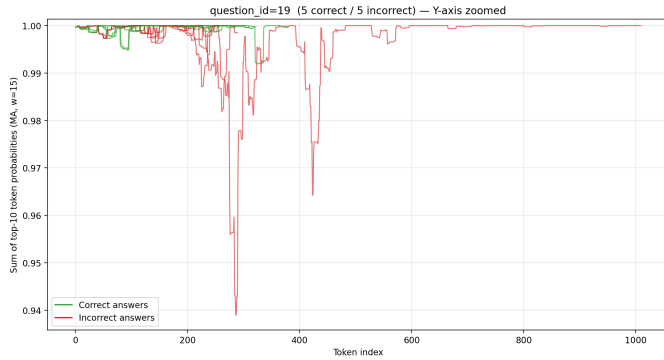


Fig. 1. Moving-average top-10 probability curves for one mixed question. Green curves are correct answers and red curves are incorrect answers. The y-axis is zoomed to expose small deviations near one.

neous: some questions show clear uncertainty dips in incorrect answers, while others show overlap or even dips in correct answers. This visual overlap already suggests that the classification task should be possible but limited.

The visual separation depends on the label balance inside the ten generated answers for a question. When roughly four or five of the ten answers are wrong, the red and green curves are usually easier to distinguish, and the red curves more often sit lower or contain deeper dips. When only one or two answers are wrong, the distinction is much less stable: a correct answer can have a red-like dip, or the few incorrect answers may not go below the green curves. Thus, the clearest qualitative evidence appears in nearly balanced mixed questions rather than in questions with only a small minority of wrong answers.

IV. FEATURE EXTRACTION AND MODELS

For every answer, features were computed from the same window-15 moving-average curve used in the plots. The raw features include mean, standard deviation, minimum, maximum, median, percentiles, linear slope, first/middle/last-third means, last-minus-first difference, location of the minimum, sharpest local drop, roughness, fractions below 0.90, 0.95, and 0.99, total uncertainty area $\sum_t (1 - s_t)$, answer length, and log length.

Absolute confidence differs strongly from one question to another. Therefore, each raw feature was also transformed within its group of ten sibling answers. For a feature value x_{iq} for answer i of question q , the relative z-score is

$$z_{iq} = \frac{x_{iq} - \mu_q}{\sigma_q + 10^{-9}}. \quad (2)$$

A within-question rank feature was also added. The final representation is the concatenation of raw, within-question z-scored, and within-question ranked features.

Five classifiers were evaluated: XGBoost, CatBoost, Gradient Boosting, Random Forest, and LightGBM. All models were trained on the 68 training questions and tested on the 64 held-out questions. Accuracy, F1 score for the correct class, and ROC-AUC were reported. The majority-class baseline on

TABLE II
CLASSIFICATION RESULTS ON 640 HELD-OUT ANSWERS.

Model	Accuracy	F1	ROC-AUC
Majority baseline	0.5250	—	—
XGBoost	0.5578	0.5928	0.5623
CatBoost	0.5750	0.6233	0.6093
Gradient Boosting	0.5734	0.6106	0.6151
Random Forest	0.5781	0.6143	0.6073
LightGBM	0.5578	0.6053	0.5875

the test set is 0.5250 because 336 of 640 test answers are correct.

V. RESULTS

Table II gives the Step 3 modeling results. All models improve over the majority baseline, but the margin is modest. Random Forest obtains the highest accuracy, 0.5781, while Gradient Boosting obtains the highest ROC-AUC, 0.6151. ROC-AUC is especially important here because the task is a confidence ranking problem: it asks whether the graph signal tends to assign higher correctness probability to correct answers even when the default 0.5 threshold is imperfect.

The class-level reports show the same trade-off. CatBoost has the highest correct-answer recall, 0.670, but only 0.470 recall for incorrect answers. Random Forest is more balanced, with 0.510 recall for incorrect answers and 0.640 recall for correct answers. Thus, the graph features are better at identifying likely-correct answers than reliably catching every incorrect one.

VI. DISCUSSION

The results support the main hypothesis, but only weakly. The clearest visual separation appears when a question has several correct and several incorrect answers, because the red curves can be compared against multiple green sibling curves. With only one or two incorrect answers, individual variation dominates: some correct answers contain deep dips because the model explores a difficult but ultimately valid reasoning step, while some incorrect answers remain highly confident until the final numeric mistake. This explains why the best AUC is in the low 0.60 range rather than close to perfect separation.

The top-10 probability sum is also a limited uncertainty measure. Because the model often assigns almost all probability mass to the top ten tokens, the signal is compressed near one. The generated arrays occasionally exceed one by a tiny amount due to floating-point and log-probability reconstruction effects, which is harmless for relative curve features but reinforces that the useful variation is very small. A sharper signal would likely include the sampled token’s own probability, full entropy over more of the vocabulary, or token-level likelihood of the complete answer.

Another limitation is dataset size and scope. The experiment did not perform prediction over the whole Turkish GSM8K dataset; it used local samples of 250 training and 250 test questions. Only 132 of these 500 sampled questions were mixed,

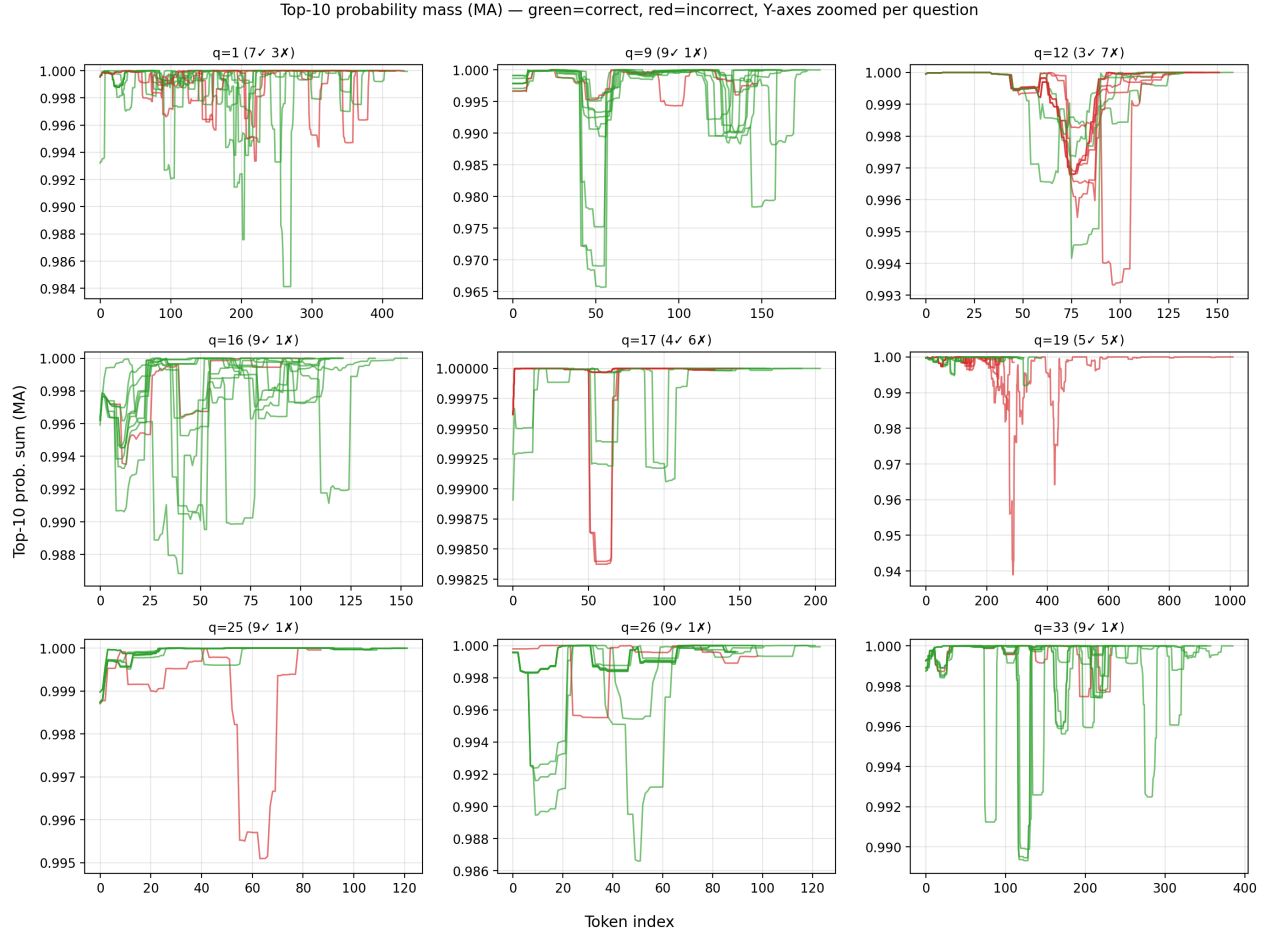


Fig. 2. Small-multiple view of nine mixed questions. Each panel has an independently zoomed y-axis. The signal is not perfectly aligned across questions, which motivates within-question relative features.

so the held-out set is meaningful but still small, especially because all evaluation is by question. Hyperparameter tuning and additional mixed questions would be the next steps before making stronger claims.

VII. CONCLUSION

This project tested whether answer correctness can be predicted from the shape of a top-10 token probability curve. On 1,320 generated answers from 132 mixed Turkish GSM8K questions, balanced mixed questions often showed clearer separation between correct and incorrect curves, while questions with only one or two wrong answers were more ambiguous. Machine-learning models trained on curve features improved over a 0.5250 majority baseline, with the best accuracy reaching 0.5781 and the best ROC-AUC reaching 0.6151. The signal is therefore real but weak: top-10 probability curves reveal some model hesitation, but they are not sufficient alone for highly reliable correctness prediction.

REFERENCES

- [1] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman, “Training verifiers to solve math word problems,” *arXiv preprint arXiv:2110.14168*, 2021.
- [2] Yildiz Technical University Computer Engineering Department Cosmos Research Group, “ytu-ce-cosmos/gsm8k_tr,” Hugging Face dataset. [Online]. Available: https://huggingface.co/datasets/ytu-ce-cosmos/gsm8k_tr. Accessed: Jun. 15, 2026.
- [3] Qwen Team, “Qwen3 technical report,” *arXiv preprint arXiv:2505.09388*, 2025.
- [4] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. Yu, J. Gonzalez, H. Zhang, and I. Stoica, “Efficient memory management for large language model serving with PagedAttention,” in *Proc. ACM SIGOPS Symp. Operating Systems Principles*, 2023.
- [5] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [6] A. V. Dorogush, V. Ershov, and A. Gulin, “CatBoost: gradient boosting with categorical features support,” *arXiv preprint arXiv:1810.11363*, 2018.
- [7] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems*, 2017.
- [8] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.